EXCELLENCE AND SERVICE

Department of Computer Science

CHRIST (Deemed to be University)

Bangalore Central Campus • Hosur Road, Bengaluru - 560029

Subject

Full Stack Development Lab

PRACTICAL ASSIGNMENT

Exercise-3: Demonstration of JavaScript Event Handling

Project: **CampusVibe — Christ University Event & Club Hub**

STUDENT CREDENTIALS

Anushya C

Register Number: **2648508**

Program: MCA / M.Sc Computer Science

1 OBJECTIVE & PROBLEM STATEMENT

Objective: To understand and implement JavaScript event handling by developing an interactive webpage that demonstrates at least three unique events and performs meaningful actions based on user interaction.

Problem Statement: Develop a responsive webpage and demonstrate the handling of a minimum of three different JavaScript events. The selected events should serve a clear purpose and improve user interaction or usability.

Evaluation Compliance Note: *Per slide specifications, the* `click` *event was excluded from the mandatory 3 unique events. This project demonstrates* ***12 unique, non-click events*** *integrated inside the* ***CampusVibe*** *domain application.*

2 IMPLEMENTED EVENTS MATRIX (12 UNIQUE NON-CLICK EVENTS)

#	EVENT NAME	TARGET ELEMENT	FUNCTIONAL PURPOSE	DYNAMIC FEEDBACK & ACTION
1	`load`	`window`	Initialize metrics and welcome user on launch	Fires native alert dialog: *"Welcome to CampusVibe — Christ University Event & Club Hub!"*
2	`input`	`#eventSearchInput`	Real-time search preview & card filtering	Updates `You typed: [query]`, character counter, and filters event cards live
3	`resize`	`window`	Detect browser width and responsive layout	Calculates exact pixel width and toggles breakpoint pill (Mobile/Tablet/Desktop)
4	`mouseover`	`.interactive-card`	Elevate card on pointer entry	Applies blue glowing ring, shadow-2xl, and upward transform (`-translate-y-2`)
5	`mouseout`	`.interactive-card`	Restore original card appearance	Reverts elevation and glowing border smoothly via transition properties
6	`focus`	`#eventSearchInput, form`	Indicate active typing focus context	Highlights input border and dynamically updates pill to `State: Active`
7	`blur`	`#eventSearchInput, form`	Sanitize and update state upon focus exit	Restores input border and dynamically updates pill to `State: Inactive`
8	`change`	`#eventTrackSelect`	Dynamic domain track switcher	Swaps track preview card with Technology, Cultural, Sports, or Entrepreneurship data
9	`submit`	`#rsvpForm`	RSVP form validation & submission	`e.preventDefault()` stops reload; validates name/email and renders success pass
10	`dblclick`	`.interactive-card`	Interactive quick star / bookmark action	Toggles `Saved` golden badge and triggers a floating toast confirmation
11	`scroll`	`window`	Track page reading progression	Calculates scroll percentage and expands top gradient bar from 0% to 100%
12	`contextmenu`	`#contextMenuDemoArea`	Custom right-click action palette	`e.preventDefault()` intercepts browser menu; renders custom floating menu at cursor

3 CAMPUS EVENTS & STUDENT CLUBS DIRECTORY

Sample Campus Events (with live search filter)

- **HackTheCampus 2026:** 24h Hackathon • SkyView Lab 4 • March 28
- **CodeVerse:** Coding Competition • Turing Lab • April 02
- **InnovateX:** Innovation & Startup Fest • Central Quad • April 08
- **Campus Beats:** Cultural Music Festival • Amphitheatre • April 15
- **TechTalks:** Future of AI Guest Lecture • Assembly Hall 1 • April 20
- **Sports Carnival:** Inter-Department Athletics • Central Ground • April 25

Sample Student Clubs (with Join Club feedback)

- **Coding Club:** Web development, open-source, and system engineering.
- **AI and Robotics Club:** Autonomous robotics, microcontrollers, and computer vision.
- **Photography Club:** Photojournalism, campus chronicles, and portfolio exhibits.
- **Music and Dance Club:** Acoustic sets, university choir, and dance crews.
- **Entrepreneurship Club:** Venture incubator, startup pitches, and founder talks.

4 GITHUB REPOSITORY LINK (DELIVERABLE 3)

OFFICIAL REPOSITORY URL:

https://github.com/Anushyaaaa/Full-Stack-Development

Branch: main • Verified & Pushed

Entry: **index.html** Script: **script.js** Lab Copy: **fsdlab2.html** Docs: **README.md**

5 CORE SOURCE CODE (VANILLA JAVASCRIPT EVENT LISTENERS)

```
// 1. Window 'load' Event
window.addEventListener('load', () => {
    updateScreenWidth();
    setTimeout(() => alert("Welcome to CampusVibe — Christ University Event & Club Hub!"), 400);
});

// 2. Input 'input' Event (Live typing & card filtering)
searchInput.addEventListener('input', (e) => {
    const q = e.target.value;
    typedValue.textContent = q || '(start typing above...)';
    charCount.textContent = q.length;
    filterCards(q.toLowerCase());
});

// 3. Window 'resize' Event (Live width & breakpoint monitor)
window.addEventListener('resize', updateScreenWidth);

// 4 & 5. 'mouseover' & 'mouseout' Events (Hover elevation & glow)
document.querySelectorAll('.interactive-card').forEach(card => {
    card.addEventListener('mouseover', () => card.classList.add('ring-2', 'ring-brand-500', '-translat
    card.addEventListener('mouseout', () => card.classList.remove('ring-2', 'ring-brand-500', '-trans
    card.addEventListener('dblclick', () => card.classList.toggle('bookmarked-card'));
});

// 6 & 7. 'focus' & 'blur' Events (Field focus state)
searchInput.addEventListener('focus', () => fieldFocusStatus.textContent = 'State: Active');
searchInput.addEventListener('blur', () => fieldFocusStatus.textContent = 'State: Inactive');

// 8. Select 'change' Event (Domain track switcher)
eventTrackSelect.addEventListener('change', (e) => renderTrackPreview(e.target.value));

// 9. Form 'submit' Event (RSVP validation & preventDefault)
rsvpForm.addEventListener('submit', (e) => {
    e.preventDefault();
    if (validateRSVP()) showConfirmation();
});

// 11. Window 'scroll' Event (0% - 100% progress indicator)
window.addEventListener('scroll', () => {
    const docHeight = document.documentElement.scrollHeight - window.innerHeight;
    const progress = docHeight > 0 ? (window.scrollY / docHeight) * 100 : 0;
    scrollBar.style.width = Math.min(100, Math.max(0, progress)) + '%';
});

// 12. Area 'contextmenu' Event (Custom right-click menu with coordinates)
contextMenuArea.addEventListener('contextmenu', (e) => {
    e.preventDefault();
    openCustomMenu(e.pageX, e.pageY);
});
```

6 AI USAGE DECLARATION (MANDATORY DELIVERABLE)

AI Tools Utilized: Google Antigravity (Powered by Google DeepMind Advanced Agentic Coding & Gemini 3.8 Flash).

Purpose of AI Assistance:

- Designing an accessible semantic HTML5 structure with responsive Tailwind CSS utilities.
- Structuring clean, decoupled vanilla JavaScript event listeners (`load` , `input` , `resize` , `mouseover` , `mouseout` , `change` , `submit` , `dblclick` , `scroll` , `contextmenu`).
- Structuring an asynchronous, non-blocking real-time event activity console.
- Formatting this academic submission document adhering to the 10-mark evaluation rubrics.

Declaration: All architectural patterns, event integrations, and UI components were reviewed, tested, and validated by the student in the local development environment.

7 REFLECTIONS & VIVA VOCE Q&A

Reflections & Learnings: Implementing the 12 events on CampusVibe reinforced the architecture of the browser's asynchronous event loop. Modern DOM event listeners using `addEventListener()` ensure complete separation of concerns between markup and logic. Custom event interception via `event.preventDefault()` was instrumental in validating the RSVP form without page reloads and implementing the coordinate-positioned custom context menu.

Q: Event Bubbling vs Event Capturing?

In Capturing, the event travels from window down to target. In Bubbling (standard default), the event executes on the target element first and bubbles upwards to ancestors.

Q: Why use event.preventDefault()?

Stops native browser behaviors like form HTTP POST page reloads or opening the OS context menu, enabling client-side validation and custom interactions.